tempssm: State-space modeling for temperature time series in R

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Summary

`tempssm` is an R package for state-space modeling of environmental temperature time series, including air and water temperature observations. It provides a practical framework for assessing how long-term trend, seasonal variation, autoregressive dependence, and optional exogenous effects contribute to observed temporal variation. The package facilitates the application of linear Gaussian state-space models estimated by Kalman filtering and smoothing, using the `KFAS` package as the computational backend (Helske, 2017).

Key Features

- Fits linear Gaussian state-space models to environmental temperature time series.
- Represents temperature dynamics using interpretable latent components: long-term trend, seasonal variation, autoregressive dependence, and optional exogenous effects.
- Supports arbitrary seasonal frequencies, while the current examples and validation focus primarily on monthly temperature data.
- Allows users to specify an arbitrary order of the autoregressive component (default: AR(1)).
- Includes time-series cross-validation tools for model evaluation.

Input Data Format

R `ts` objects are the primary input format for `tempssm`. The temperature series passed to `tempssm()` should be supplied as a univariate `ts` object, and optional exogenous variables can be supplied as univariate or multivariate `ts` objects. The `ts` class is base R's standard format for regularly spaced time series (see the `stats::ts` documentation: https://search.r-project.org/R/refmans/stats/html/ts.html). The package also provides utility functions for converting common tabular and observational data formats into `ts` objects before model fitting.

The seasonal cycle used by `tempssm()` is taken from the `frequency` attribute of the input `ts` object. For example, `frequency = 12` represents monthly data.

Prior Art and Scope

`tempssm` provides a domain-focused workflow for analyzing temperature time series with linear Gaussian state-space models. It brings together model construction, component extraction, uncertainty summaries, residual diagnostics, visualization, and time-series cross-validation in a single R package interface tailored to temperature applications.

The package builds on established statistical methodology, including linear Gaussian state-space modeling, Kalman filtering, and Kalman smoothing. Model estimation is handled through the `KFAS` package, which provides a general framework for state-space models in R.

The initial implementation was adapted from the supplementary code provided by Baba (2024), accompanying Baba et al. (2024), which analyzed sea temperature trends using a linear Gaussian state-space model. The supplementary code is publicly available at:

https://github.com/logics-of-blue/sea-temperature-trend-jogashima

Compared with that prior implementation, `tempssm` extends the workflow into a reusable R package interface with input validation, documented S3 methods, tests, diagnostics, cross-validation utilities, and examples for broader temperature time-series analysis.

Model Overview

The model implemented in `tempssm` can be viewed as an extension of the Basic Structural Time Series Model (BSTSM), a standard state-space formulation that represents an observed time series using latent trend, seasonal, and irregular components. Following the temperature time-series application of Baba et al. (2024), `tempssm` keeps this interpretable decomposition and adds autoregressive dependence and optional exogenous effects. This makes it useful for separating long-term temperature change, seasonal cycles, and short-term departures.

The model is estimated with `KFAS`, and `tempssm()` returns both filtering and smoothing estimates. Unless otherwise stated, the summaries, diagnostics, and plots in this vignette use smoothed state estimates.

The full mathematical specification, including the observation equation, state decomposition, seasonal constraint, autoregressive component, exogenous component, parameter count, and estimation procedure, is described in the separate model specification document, available as a PDF in the package repository:

https://github.com/akihirao/tempssm/blob/main/vignettes/model-specification.pdf

Setup

Load `tempssm` before running the examples below.

```r
## Set libraries
library(tempssm)
```

The objective of this practice is to demonstrate the basic application of a linear Gaussian state-space model to a univariate temperature time series. This example introduces the basic workflow for fitting a `tempssm` model, inspecting the summary output, plotting latent components, checking residual diagnostics, and making short-term predictions.

Example Dataset

This tutorial uses `temp_MtFuji`, a long-term monthly air temperature time series observed at the summit of Mt. Fuji, Japan.

- **Dataset**: Monthly mean air temperature at the summit of Mt. Fuji, Japan
- **Format**: Univariate `ts` object
- **Frequency**: 12 (monthly)
- **Unit**: Degrees Celsius
- **Period**: July 1932 to June 2026

The dataset was created by processing content from the Japan Meteorological Agency (JMA), Past Weather Data Download page: https://www.data.jma.go.jp/risk/obsdl/index.php. The series may contain missing values from the source data; values with JMA quality flags below 8, where 8 indicates a normal value with no quality problem, are also treated as missing values.

```r
data(temp_MtFuji) # load a ts object of temperature at Mt. Fuji
head(temp_MtFuji)
```

```
##        Jul   Aug   Sep   Oct   Nov   Dec
## 1932   5.5   5.7   1.8  -4.6  -9.5 -12.9
```

```r
summary(temp_MtFuji)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs
##  -25.20  -14.60   -5.80   -6.36    2.00    8.30      11
```

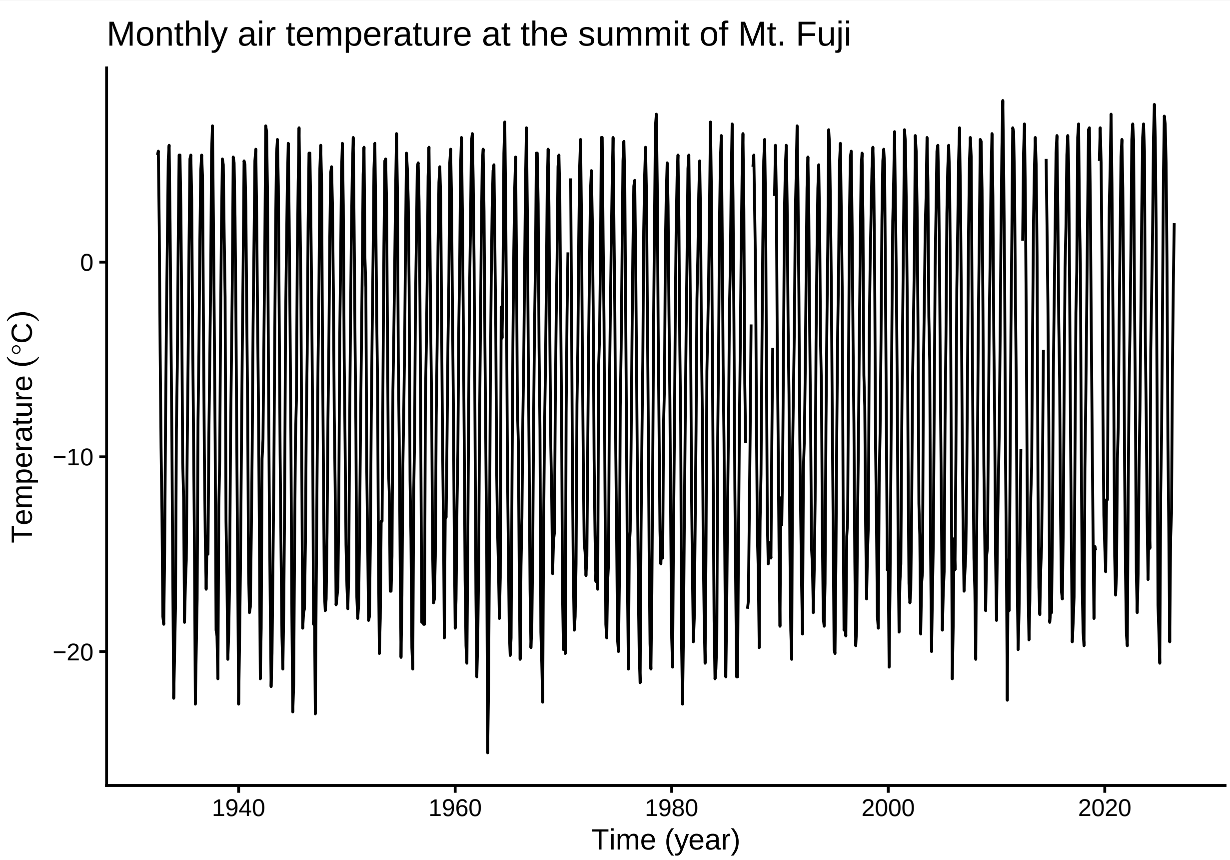
This dataset contains missing observations. `tempssm()` can retain missing values in the response temperature series and treat them as unobserved responses during Kalman filtering and smoothing.

Plot the Time Series

We begin by visualizing the monthly air temperature time series to examine its overall structure, including apparent trends, seasonal variability, and whether missing observations are present.

```r
plt_mtfuji_temp <- forecast::autoplot(temp_MtFuji) +
  ggplot2::labs(
    y = expression(Temperature ~ (degree * C)),
    x = "Time (year)"
  ) +
  ggplot2::ggtitle("Monthly air temperature at the summit of Mt. Fuji") +
  ggplot2::theme_classic()

plot(plt_mtfuji_temp)
```



The overall mean air temperature is approximately -6.4 °C, and a clear annual seasonal pattern is visible. The series contains 11 missing observations. Although the observed temperature appears to increase gradually over the long term, year-to-year variation and strong seasonality make the underlying trend difficult to assess from the raw time series alone.

Fit a State-Space Model

When a `ts` object containing temperature time-series data (here, `temp_MtFuji`) is passed to the core function `tempssm()`, model construction and parameter estimation are performed together. The returned S3 object of class `tempssm` (here, `res`) stores the filtering and smoothing estimates, as well as the constructed model and input data. By default, `tempssm()` fits a first-order autoregressive model.

```
# model with first-order autoregressive component
res <- tempssm(temp_MtFuji) # AR(1), the default model
summary(res)
```

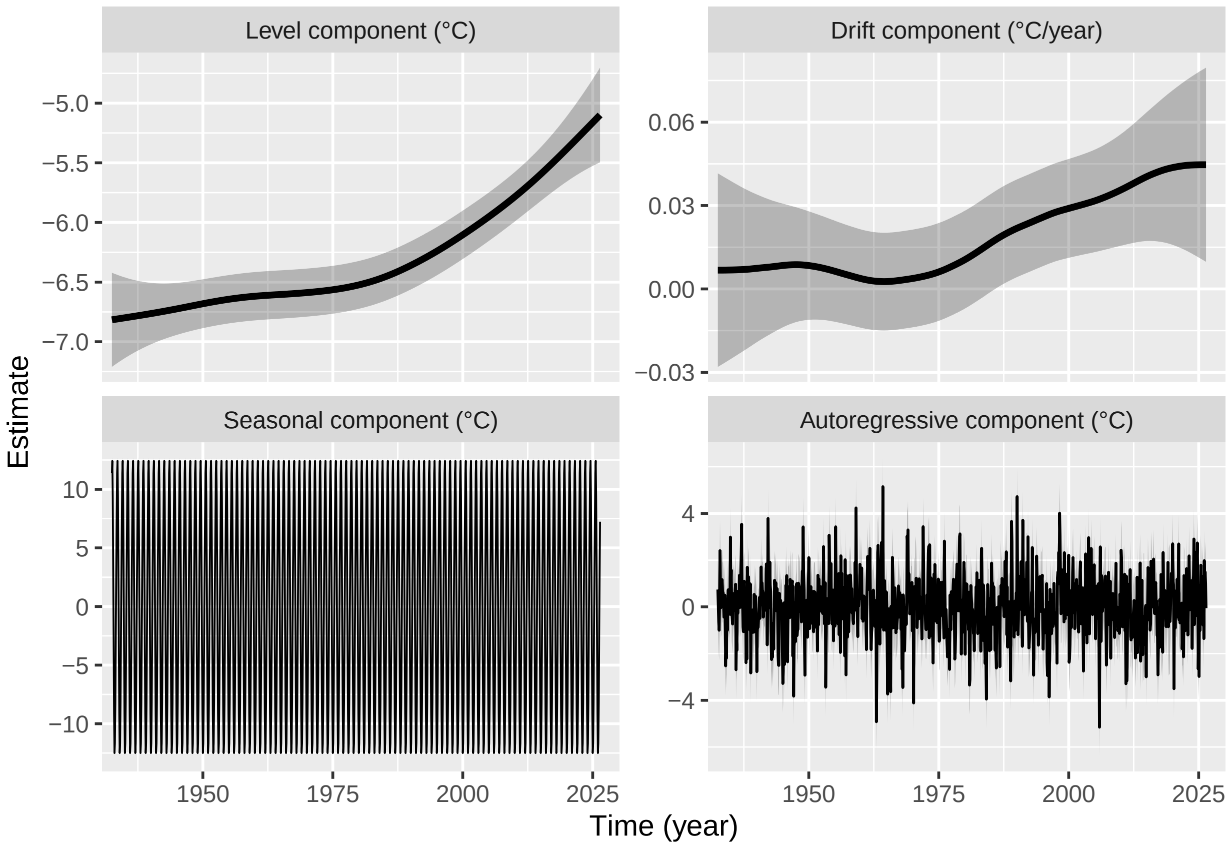
```
## tempssm summary
## -----------------
## Call:
## tempssm(temp_data = temp_MtFuji)
##
## Model fit:
##   Likelihood type: marginal
##   Log-likelihood : -2055.83
##   k              : 5
##   Diffuse states : 13
##   Converged      : TRUE
##
## Variance parameters:
##   Observation (H): 0.4919416
##   State (Q trend): 1.146231e-08
##   State (Q season): 4.153786e-24
##   State (Q ar): 1.901346
##
## Components of auto-regression:
##   Order of AR: 1
##   Coefficient of AR1: 0.2564485
```

First, confirm from the summary output that the model has converged (`Converged: TRUE`). The summary also reports the log-likelihood, parameter count (`k`), likelihood type, and number of diffuse initial states. The estimated parameters include the observation error variance (`H`) and the process error variances for the long-term trend (`Q trend`), seasonal component (`Q season`), and autoregressive component (`Q ar`), as well as the autoregressive coefficient (`AR1` in the default model). See the detailed manual for extracting and using these quantities directly; its location is listed at the end of this quick tutorial.

Plot Model Components

We visualize the estimated long-term evolution of temperature levels and their rates of change (drift) by extracting the corresponding latent components from the state-space model. Additionally, seasonal variability and autoregressive dependence are plotted out, allowing the underlying trend behavior to be examined more clearly.

```
# plot all components at once
plot(res)
```



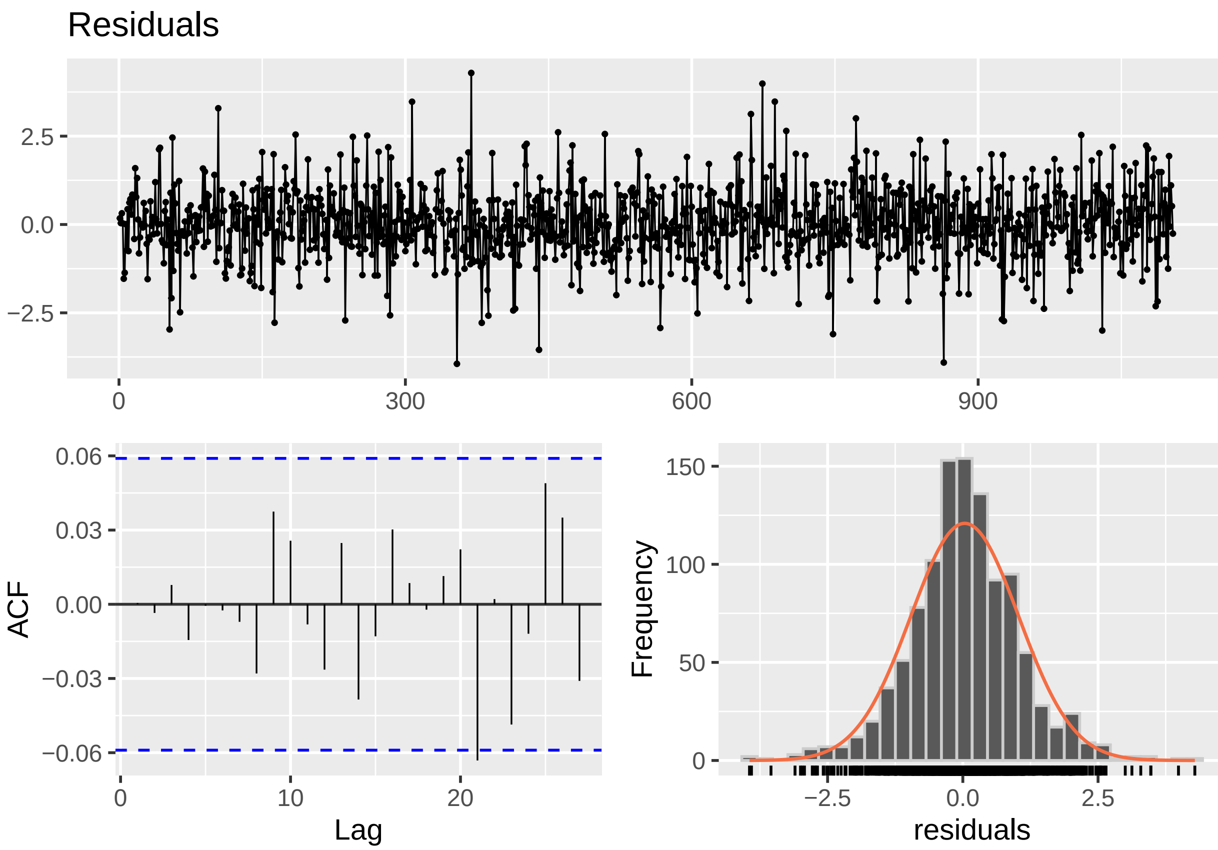
The level component summarizes gradual changes in the underlying temperature level after removing the dominant seasonal pattern. In this example, the estimated long-term level indicates a gradual increase in temperature over the study period. The drift component is positive on average but small relative to the seasonal variation. The seasonal component captures the recurring annual cycle, while the autoregressive component represents shorter-term departures from the trend and seasonal pattern. The shaded gray areas represent 95% confidence intervals for the estimated latent states, illustrating the uncertainty associated with each estimated component.

The standard plotting interface is `plot(res)`. The explicit helper `plot_tempssm_components(res)` produces the same component plot and can be useful in scripts where a descriptive function name is preferred. The ggplot2-style S3 interface `ggplot2::autoplot(res)` is also available. These interfaces return a faceted `ggplot` object, so selected components can be stored and customized with standard ggplot2 layers, for example, `plot_tempssm_components(res, component = c("level", "drift")) + ggplot2::theme_bw()`.

Check Residual Diagnostics

The package provides diagnostic tools for checking whether the fitted model has left notable structure in the residuals. In particular, residual time-series plots, residual autocorrelation, residual distributions, and Ljung-Box tests can be used to assess remaining temporal dependence and departures from the Gaussian error assumption.

```r
r <- get_tempssm_residuals(res)
plot_tempssm_residuals(r, frequency = frequency(res$temp_data))
```




```r
diag <- diagnose_residuals(res)
diag
```

```
## # A tibble: 1 x 4
##   lb_stat lb_lag lb_pvalue kurtosis
##     <dbl>  <dbl>     <dbl>    <dbl>
## 1    4.40     12     0.975     4.39
```

Here, `get_tempssm_residuals()` extracts standardized recursive residuals. `plot_tempssm_residuals()` displays the residual time series in the upper panel, the ACF plot in the lower-left panel, and the residual frequency distribution in the lower-right panel. `diagnose_residuals()` returns a compact table with the Ljung-Box test and residual kurtosis. For monthly data, the default Ljung-Box lag is 12. These plots and the test result should be checked for any notable residual patterns. In this example, the Ljung-Box test indicated no significant residual autocorrelation up to lag 12 ($P > 0.05$).

For more detailed discussion of residual diagnostics, see the detailed manual.

Extract Components

The fitted object stores the smoothed latent states in `res$kfs$alphahat`. Each column corresponds to one state component in the state-space model, such as the level, drift, seasonal states, and autoregressive state. Looking at the first few rows is useful for understanding how the model output is organized.

```r
# Smoothing estimates
alpha_hat <- res$kfs$alphahat
head(alpha_hat)
```

```
##                level        slope sea_dummy1 sea_dummy2 sea_dummy3 sea_dummy4
```

```
## Jul 1932 -6.816007 0.0005640191  11.386172   7.235620   2.725050  -2.300199
## Aug 1932 -6.815443 0.0005640234  12.425581  11.386172   7.235620   2.725050
## Sep 1932 -6.814879 0.0005640325   9.370378  12.425581  11.386172   7.235620
## Oct 1932 -6.814315 0.0005640435   3.464614   9.370378  12.425581  11.386172
## Nov 1932 -6.813751 0.0005640508  -2.878391   3.464614   9.370378  12.425581
## Dec 1932 -6.813187 0.0005640535  -9.076789  -2.878391   3.464614   9.370378
##          sea_dummy5 sea_dummy6 sea_dummy7 sea_dummy8 sea_dummy9 sea_dummy10
## Jul 1932  -8.183394 -11.668060 -12.500582  -9.076789  -2.878391    3.464614
## Aug 1932  -2.300199  -8.183394 -11.668060 -12.500582  -9.076789   -2.878391
## Sep 1932   2.725050  -2.300199  -8.183394 -11.668060 -12.500582   -9.076789
## Oct 1932   7.235620   2.725050  -2.300199  -8.183394 -11.668060  -12.500582
## Nov 1932  11.386172   7.235620   2.725050  -2.300199  -8.183394  -11.668060
## Dec 1932  12.425581  11.386172   7.235620   2.725050  -2.300199   -8.183394
##          sea_dummy11      arima1
## Jul 1932    9.370378  0.74270058
## Aug 1932    3.464614  0.07576107
## Sep 1932   -2.878391 -0.64035913
## Oct 1932   -9.076789 -1.00172111
## Nov 1932  -12.500582  0.22370694
## Dec 1932  -11.668060  2.40713430
```

For routine use, helper functions provide a simpler way to extract individual components as `ts` objects with the original time index. The level component represents the estimated long-term temperature level, while the drift component represents its rate of change per year.

```r
# Smoothing estimate of level component
level_ts <- get_level_ts(res)

# Smoothing estimate of drift component
drift_ts <- get_drift_ts(res)

# Average drift rate per year across the full period
mean_drift_year <- mean(drift_ts)
mean_drift_year
```

```
## [1] 0.0183029
```

Average annual change in air temperature is approximately 0.02 °C. This value is the mean of the estimated drift component over the full observation period, and can be read as a model-based summary of the average long-term rate of temperature change.

Make Short-Term Predictions

A fitted `tempssm` object can also be passed to `predict()` to obtain short-term predictions. By default, `predict(res)` returns a one-step-ahead prediction beyond the end of the observed series. This is useful for visual checks of how the fitted model extrapolates the estimated level, seasonal, and autoregressive components.

```r
pred_1 <- predict(res)
pred_1
```

```
##           Jul
## 2026 6.276348
```

Predictions for multiple future time points can be requested by setting the `n.ahead` argument.

```r
pred_12 <- predict(res, n.ahead = 12)
pred_12
```

```
##            Jan        Feb        Mar        Apr        May        Jun
## 2026
## 2027 -17.573776 -16.737527 -13.249136  -7.362216  -2.333243   2.181051
##            Jul        Aug        Sep        Oct        Nov        Dec
## 2026   6.276348   7.330106   4.281351  -1.619989  -7.959091 -14.153719
## 2027
```

These predictions should be interpreted as model-based extrapolations rather than definitive forecasts. Uncertainty generally increases as the prediction horizon becomes longer, and long-horizon predictions can be sensitive to model assumptions about trend, seasonality, and autoregressive dependence.

Read Your Own CSV Data

For monthly temperature data stored in a CSV file, prepare columns named `Year`, `Month`, and `Temp`. Use `NA` for missing temperature values, and keep the corresponding `Year` and `Month` entries.

```
Year,Month,Temp
2010,8,13.6
2010,9,6.8
2010,10,NA
2010,11,-1.4
...
```

The CSV file should be comma-separated and UTF-8 encoded. The package includes an example CSV file in `inst/extdata`. The helper function `read_monthly_temp_ts()` reads this type of CSV file and converts it into an R `ts` object for use with `tempssm()`.

```r
path <- system.file(
  "extdata",
  "example_monthly_temp.csv",
  package = "tempssm"
)

csv_temp <- tempssm::read_monthly_temp_ts(path)
head(csv_temp)
```

```
##        Jan Feb Mar Apr May Jun Jul   Aug   Sep   Oct   Nov   Dec
## 2010                                13.6   6.8   0.2  -6.8 -12.5
## 2011 -18.8
```

Further Reading

The examples above illustrate the basic univariate workflow for fitting, diagnosing, visualizing, and making short-term predictions from a `tempssm` model. The detailed manual extends this workflow to exogenous variables, additional model diagnostics, and time-series cross-validation:

https://github.com/akihirao/tempssm/blob/main/tools/manual/tempssm_manual.pdf